

6	(a)	oil drop charged by friction/beta source	B1	
		between parallel <u>metal</u> plates	B1	
		plates are horizontal	(1)	
		adjustable potential difference/field between plates	B1	
		until oil drop is stationary	B1	
		$mg = q \times V/d$	B1	
		symbols explained	(1)	
		oil drop viewed through microscope	(1)	
		m determined from terminal speed of drop (when p.d. is zero)	(1)	
		<i>(any two extras, 1 each)</i>	B2	[7]
	(b)	$3.2 \times 10^{-19} \text{ C}$	A1	[1]
7	(a)	minimum energy to remove an electron from the metal/surface	B1	[1]